SANCHAYI SHARMA

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SUMMARY

Software Engineer with an M.S. in Computer Science and experience building backend APIs, full-stack products, and AI-enabled tools. Improved API latency by 40% and reduced manual data preparation by 60% through database and data-pipeline optimization.

PROFESSIONAL EXPERIENCE

UNIVERSITY OF TEXAS AT ARLINGTON | Graduate Research Assistant Aug 2025 – May 2026

- Replaced fragmented spreadsheet and script workflows with a React and FastAPI dashboard for tracking research pipelines, processing datasets, and reviewing model predictions.
- Built Python and PostgreSQL ingestion pipelines for 5+ sources with schema checks and transformations, reducing manual data preparation by 60%.
- Exposed a Hugging Face classifier through asynchronous FastAPI endpoints with Pydantic validation, structured error handling, and pytest-based tests; the model achieved 88% held-out accuracy.

NEXAMUSE | Full-Stack Software Engineer Aug 2023 – Jul 2024

- Delivered React, TypeScript, and Node.js features for onboarding, content delivery, and progress tracking on a web and mobile platform serving 5,000+ active users.
- Identified an N+1 query bottleneck under concurrent traffic; restructured SQL queries and added Redis caching to reduce average API latency by 40%.
- Consolidated duplicated UI logic into reusable React components and centralized Redux state across core product workflows.
- Implemented JWT access and refresh-token authentication and raised Jest unit and integration test coverage from 40% to 65%.

ACV INCORPORATION | Software Engineering Intern Jan 2023 – Jul 2023

- Built Node.js and MySQL billing APIs with request validation and idempotency checks to prevent duplicate transactions on retried requests.
- Increased automated payment and refund test coverage to 70%+, uncovering a refund-calculation defect before staging.

SELECTED PROJECTS

AI-Powered PR Risk Assistant [GitHub] | FastAPI, React, PostgreSQL, GitHub API, LLM

- Built a GitHub webhook-based code-review service that retrieves pull-request changes and generates structured findings about potential bugs, security risks, and missing tests.
- Added asynchronous processing, duplicate-event checks, and PostgreSQL persistence, with a React dashboard for reviewing findings and pull-request history.

TraceOps: Incident Root-Cause Analysis [GitHub] | Python, FastAPI, OpenTelemetry, Prometheus, Docker

- Built an incident investigation platform that correlates distributed traces, logs, and metrics to pinpoint where failures originate in a microservice chain, correctly identifying the root cause in all 4 injected failure scenarios (database outage, slow queries, timeouts, service crash).
- Instrumented 3 FastAPI services with OpenTelemetry and Prometheus alerting; alerts open a single correlated incident with an evidence-backed report in a React/TypeScript dashboard, validated by 38 pytest tests.
- Diagnosed a timeout-ordering bug that made upstream services blame the wrong dependency during database outages, and fixed a trace-ID mismatch that prevented joining logs to traces.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C

Backend & APIs: FastAPI, Node.js, Express, REST APIs

Frontend: React, Redux, Next.js

Databases & Cloud: PostgreSQL, MySQL, Redis, AWS (EC2, S3, Lambda), Docker

AI & Developer Tools: LLM APIs, Hugging Face, Git, GitHub Actions, Jest, pytest

EDUCATION

UNIVERSITY OF TEXAS AT ARLINGTON | M.S. Computer Science, GPA: 3.7/4.0 Aug 2024 – May 2026

GUJARAT TECHNOLOGICAL UNIVERSITY | B.E. Computer Engineering Aug 2020 – May 2024

CERTIFICATIONS

AWS Certified Developer – Associate (DVA-C02) | Data Structures & Algorithms – UCSD (Coursera)